

# The Foundations: Logic and Proofs

Chapter 2, Part II: Predicate Logic

# Predicates and Quantifiers

Section 1.4

# What is a Predicate?

**Predicate = statement with a variable**

A **predicate** is a statement that depends on a variable and becomes **true or false** **once the variable is given a value.**

**Example:**

- $P(x)$ :  $x$  is even
- $P(x)$ :  $x > 3$
- $P(x)$ :  $|x| = 1$

By itself, it is **not true or false**.

It becomes true or false **after you put a value of  $x$  or a quantifier.**

# Why We Need Predicate Logic?

## Problem with Propositional Logic:

Propositional logic only deals with whole statements (propositions) being true or false. But many arguments involve **objects and their properties**:

### Example:

#### Statement:

1. All birds can fly.
2. A sparrow is a bird.

#### Therefore:

➡ A sparrow can fly.

In propositional logic, these would be  $p, q, r$  with no connection. We can't express "all" or "some" relationships.

**Solution:** Predicate logic adds variables, predicates, and quantifiers.

### Statement 2:

1. All students must register for courses.
2. Ali is a student.

#### Therefore:

➡ Ali must register for courses.

## Now write it in Predicate Logic

### **Predicates Example1:** **Predicates Example2:**

- $B(x)$ : x is a bird
- $F(x)$ : x can fly

- $S(x)$ : x is a student
- $R(x)$ : x must register

### **Logical form:**

1.  $\forall x(B(x) \rightarrow F(x))$
2.  $B(\text{sparrow})$

### **Conclusion:**

$F(\text{sparrow})$

### **Logical form:**

1.  $\forall x(S(x) \rightarrow R(x))$
2.  $S(\text{Ali})$

### **Conclusion:**

$R(\text{Ali})$

# Variables and Domain

**Variables:**  $x, y, z$  (placeholders for objects)

**Domain:** The domain tells us which values we are allowed to test.

□ **Domain is denoted by  $U$ .**

**Example:**

**Domain  $D = \mathbb{Z}$  (all integers)**

**•  $D = \{1, 2, 3, 4, 5\}$**

- Always specify the domain - it **changes the meaning** of the predicate.
- (Changing the domain changes the answer.)

# Propositional Functions

Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the domain (or bound by a quantifier)

Same predicate, different domain  $\rightarrow$  different meaning

**Example:**

$P(x)$ : “ $x > 0$ ”

- $U =$  integers  $\rightarrow \forall x P(x)$  is **False**
- $U =$  positive integers  $\rightarrow \forall x P(x)$  is **True**

**Always have to specify domain( $U$ )**

# Examples of Propositional Functions

- Let “ $x + y = z$ ” be denoted by  $R(x, y, z)$  and  $U$  (for all three variables) be the integers. Find these truth values:
  - $R(2, -1, 5)$  **Solution: F**
  - $R(3, 4, 7)$  **Solution: T**
  - $R(x, 3, z)$  **Solution: Not a Proposition**
- Now let “ $x - y = z$ ” be denoted by  $Q(x, y, z)$ , with  $U$  as the integers. Find these truth values:
  - $Q(2, -1, 3)$  **Solution: T**
  - $Q(3, 4, 7)$  **Solution: F**
  - $Q(x, 3, z)$  **Solution: Not a Proposition**



# What is a Truth Set?

Given a predicate  $P(x)$  and a domain  $D$ , the truth set of  $P(x)$  is the set of **all elements in  $D$  for which  $P(x)$  is true** .

$$\{x \in D \mid P(x) \text{ is true}\}$$

## EXAMPLE

**Predicate:**

$$P(x): x > 0$$

**Domain:**

$$D = \{-2, -1, 0, 1, 2, 3\}$$

**Solution:**

**Truth set:  $\{1, 2, 3\}$**

| <b>x</b>  | <b><math>x &gt; 0</math></b> | <b>True/False</b> |
|-----------|------------------------------|-------------------|
| <b>-2</b> | $-2 > 0$                     | False             |
| <b>-1</b> | $-1 > 0$                     | False             |
| <b>0</b>  | $0 > 0$                      | False             |
| <b>1</b>  | $1 > 0$                      | True              |
| <b>2</b>  | $2 > 0$                      | True              |
| <b>3</b>  | $3 > 0$                      | True              |

## Predicate:

$$P(x): |x| = 1$$

## Domain:

Integers

## Test values:

- $x = -1 \rightarrow |-1| = 1 \rightarrow \text{TRUE}$
- $x = 1 \rightarrow |1| = 1 \rightarrow \text{TRUE}$
- Others  $\rightarrow \text{FALSE}$

## Truth set:

$\{-1, 1\}$

# Compound Expressions

- Connectives from propositional logic carry over to predicate logic.
- If  $P(x)$  denotes “ $x > 0$ ,” find these truth values:

1.  $P(3) \vee P(-1)$       **Solution: T**

1.  $P(3): 3 > 0 \rightarrow \text{True (T)}$

2.  $P(-1) : -1 > 0 \rightarrow \text{False (F)}$

2.  $P(3) \wedge P(-1)$       **Solution: F**

3.  $P(3) \rightarrow P(-1)$       **Solution: F**

4.  $P(3) \rightarrow \neg P(-1)$       **Solution: T**

**$T \vee F = T$**

- Expressions with variables are not propositions and therefore **do not have truth values**. For example,

$$P(3) \wedge P(y)$$

$$P(x) \rightarrow P(y)$$

# Quantifiers

## 1. Universal quantifier ( $\forall$ )

$\forall x \in D, P(x)$  is TRUE

“For **every**  $x$  in the domain,  $P(x)$  is true”

Rule using truth set:

Truth set = whole domain

## 2. Existential quantifier ( $\exists$ )

$\exists x \in D, P(x)$  is TRUE

“There **exists at least one**  $x$  in the domain where  $P(x)$  is true”

Rule using truth set:

Truth set is NOT empty

### Example 1:

Domain =  $\{1,2,3\}$

Predicate:  $x > 0$

Truth set =  $\{1,2,3\}$

✓  $\forall x P(x)$  is TRUE

### Example 2:

Domain =  $\{1,2,3\}$

Predicate:  $x$  is even

Truth set =  $\{2\}$

✓  $\exists x P(x)$  is TRUE

### Example 3:

Domain  $D = \{0,1,2\}$

Predicate  $P(x): x^2 = 1$

Truth set =  $\{1\}$

✓  $\exists x P(x)$  is TRUE

**Truth set** = all values that make the predicate true.

$\forall$  means the truth set is the whole domain;

$\exists$  means it's non-empty.

# How Quantifiers Work?

Imagine your domain has 3 objects: {Alice, Bob, Charlie}

- $S(x)$ : "x is a student"

**Universal ( $\forall$ ) = AND over all objects:**

$\forall x S(x)$  means:

$S(\text{Alice})$  **AND**  $S(\text{Bob})$  **AND**  $S(\text{Charlie})$

(All three must be true)

**Existential ( $\exists$ ) = OR over all objects:**

$\exists x S(x)$  means:

$S(\text{Alice})$  **OR**  $S(\text{Bob})$  **OR**  $S(\text{Charlie})$

**Domain:**

$$D = \{-3, -2, -1, 0, 1, 2, 3\}$$

**Predicate:**

$$P(x): x^2 < 5$$

1. Find the truth set of  $P(x)$

2. Determine whether:

$\exists x P(x)$  is true or false

$\forall x P(x)$  is true or false

**Domain:**

$$D = \{-3, -2, -1, 0, 1, 2, 3\}$$

**Predicate:**

$$P(x): x^2 < 5$$

1. Find the truth set of  $P(x)$

2. Determine whether:

$\exists x P(x)$  is true or false

$\forall x P(x)$  is true or false

**SOLUTION:**

- **Truth set:**  $\{-2, -1, 0, 1, 2\}$
- **$\exists x P(x)$ :** TRUE



**Domain:**

$$D = \{1, 2, 3, 4, 5\}$$

**Predicate:**

$P(x)$ :  $x$  is a prime number

1. Find the truth set of  $P(x)$
2. Determine whether:
  - $\exists x P(x)$  is true or false
  - $\forall x P(x)$  is true or false

**Domain:**

$$D = \{1, 2, 3, 4, 5\}$$

**Predicate:**

$P(x)$ :  $x$  is a prime number

1. Find the truth set of  $P(x)$

2. Determine whether:

$\exists x P(x)$  is true or false

$\forall x P(x)$  is true or false

**SOLUTION:**

- **Truth set:**  $\{2, 3, 5\}$
- **$\exists x P(x)$ :** TRUE
- **$\forall x P(x)$ :** FALSE (1 and 4 are not prime)

# Properties of Quantifiers

- The truth value of  $\exists x P(x)$  and  $\forall x P(x)$  depend on both the **propositional function  $P(x)$  and on the domain  $U$ .**
- **Examples:**
  1. If  $U$  is the positive integers and  $P(x)$  is the statement “ $x < 2$ ”, then  $\exists x P(x)$  is true, but  $\forall x P(x)$  is false.
  2. If  $U$  is the negative integers and  $P(x)$  is the statement “ $x < 2$ ”, then both  $\exists x P(x)$  and  $\forall x P(x)$  are true.
  3. If  $U$  consists of 3, 4, and 5, and  $P(x)$  is the statement “ $x > 2$ ”, then both  $\exists x P(x)$  and  $\forall x P(x)$  are true. But if  $P(x)$  is the statement “ $x < 2$ ”, then both  $\exists x P(x)$  and  $\forall x P(x)$  are false.

# Precedence of Quantifiers

- The quantifiers  $\forall$  and  $\exists$  have higher precedence than all the logical operators.
- For example,  $\forall x P(x) \vee Q(x)$  means  $(\forall x P(x)) \vee Q(x)$
- $\forall x (P(x) \vee Q(x))$  means something different.
- Unfortunately, often people write  $\forall x P(x) \vee Q(x)$  when they mean  $\forall x (P(x) \vee Q(x))$ .

# Translating from English to Logic

**Example 1:** Translate the following sentence into predicate logic: **“Every student in this class has taken a course in Java.”**

**Solution:** First decide on the domain  $U$ .

**Option A:** Domain = students in this class only.

Let  $J(x)$  = "x has taken Java."

We just say:  $\forall x J(x)$

**Meaning:** *Every person in this domain has taken Java.*

**Option B:** Domain = all people.

Let  $S(x)$  = "x is a student in this class"

$J(x)$  = "x has taken Java."

Right way:  $\forall x (S(x) \rightarrow J(x))$

**Meaning:** *For all people, if they are in this class, then they have taken Java.*

$\forall x (S(x) \wedge J(x))$  is not correct. What does it mean?

# Translating from English to Logic

**Example 1:** Translate the following sentence into predicate logic: **“Some student in this class has taken a course in Java.”**

**Solution:** First decide on the domain  $U$ .

**Option A:** Domain = students in this class only.

Let  $J(x)$  = "x has taken Java."

We just say:  $\exists x J(x)$

*Meaning: There exists at least one student in the domain who has taken Java.*

**Option B:** Domain = all people.

Let  $S(x)$  = "x is a student in this class"

$J(x)$  = "x has taken Java."

Right way:  $\exists x (S(x) \wedge J(x))$

*Meaning: There exists a person who is in this class(student) and has taken Java.*

**$\exists x S(x) \wedge J(x)$  is not correct. What does it mean?**

# Equivalences in Predicate Logic

- Two logic statements are equivalent if they are **always true under the same conditions** (any predicates, any domain).
- The notation  $S \equiv T$  indicates that  $S$  and  $T$  are logically equivalent.
- **Example:**  $\forall x \neg\neg S(x) \equiv \forall x S(x)$

# Thinking about Quantifiers as Conjunctions and Disjunctions

- If the domain is finite, a **universally quantified proposition** is equivalent to a conjunction of propositions without quantifiers and an **existentially quantified proposition** is equivalent to a disjunction of propositions without quantifiers.
- **Example:**
- **Domain = {1, 2, 3}**
  - $\forall x P(x) = P(1) \wedge P(2) \wedge P(3)$  (*All must be true*)
  - $\exists x P(x) = P(1) \vee P(2) \vee P(3)$  (*At least one true*)



# Negating Quantified Expressions

## Negating "All"

- **Original:** "Every student has taken Java" =  $\forall x J(x)$
- **Negation:** "Not every student..." =  $\neg \forall x J(x)$
- **Meaning:** "There exists a student who has NOT taken Java" =  $\exists x \neg J(x)$

So:  $\neg \forall x J(x) \equiv \exists x \neg J(x)$

## Negating "Some"

- **Original:** "Some student has taken Java" =  $\exists x J(x)$
- **Negation:** "No student has taken Java" =  $\neg \exists x J(x)$
- **Meaning:** "Every student has NOT taken Java" =  $\forall x \neg J(x)$

So:  $\neg \exists x J(x) \equiv \forall x \neg J(x)$

# De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are:

| TABLE 2 De Morgan's Laws for Quantifiers. |                             |                                            |                                           |
|-------------------------------------------|-----------------------------|--------------------------------------------|-------------------------------------------|
| <i>Negation</i>                           | <i>Equivalent Statement</i> | <i>When Is Negation True?</i>              | <i>When False?</i>                        |
| $\neg\exists x P(x)$                      | $\forall x\neg P(x)$        | For every $x$ , $P(x)$ is false.           | There is an $x$ for which $P(x)$ is true. |
| $\neg\forall x P(x)$                      | $\exists x\neg P(x)$        | There is an $x$ for which $P(x)$ is false. | $P(x)$ is true for every $x$ .            |

- The reasoning in the table shows that:

$$\neg\forall x P(x) \equiv \exists x\neg P(x)$$

$$\neg\exists x P(x) \equiv \forall x\neg P(x)$$

- These are important. You will use these.

$$1. \neg \forall x J(x) \equiv \exists x \neg J(x)$$

**Domain:** {Alice, Bob, Charlie}

● = knows Java ( $J(x)$ )

○ = does NOT know Java ( $\neg J(x)$ )

Alice Bob Charlie

●   ○   ●

**SOLUTION:**

$\neg \forall x J(x) \rightarrow$  "Not everyone knows Java"  $\rightarrow$  **true** because Bob does NOT know Java

$\exists x \neg J(x) \rightarrow$  "There exists someone who does not know Java"  $\rightarrow$  **true** because Bob is that someone

## 2. $\neg \exists x J(x) \equiv \forall x \neg J(x)$

**Domain:** {Alice, Bob, Charlie}

● = knows Java ( $J(x)$ )

○ = does NOT know Java ( $\neg J(x)$ )

Alice Bob Charlie

○ ○ ○

**SOLUTION:**

$\neg \exists x J(x) \rightarrow$  "There does NOT exist anyone who knows Java"  $\rightarrow$  **true** because no one is shaded

$\forall x \neg J(x) \rightarrow$  "Everyone does not know Java"  $\rightarrow$  **true** because all are unshaded

# Translation from English to Logic

## Examples:

1. “Some student in this class has visited Mexico.”

**Solution:** Let  $M(x)$  denote “ $x$  has visited Mexico” and  $S(x)$  denote “ $x$  is a student in this class,” and  $U$  be all people.

$$\exists x (S(x) \wedge M(x))$$

2. “Every student in this class has visited Canada or Mexico.”

**Solution:** Add  $C(x)$  denoting “ $x$  has visited Canada.”

$$\forall x (S(x) \rightarrow (M(x) \vee C(x)))$$

# Translating from English into Logical Expressions

- **Domain:  $U = \{\text{Alice, Bob, Charlie}\}$**
- **Let's use:**
  - **$F(x)$ : "x is a student"**
  - **$J(x)$ : "x knows Java"**
  - **Solve:**
    - 1: "Everything is a student"**
    - 2: "Nothing is a student"**
    - 3: "All students know Java"**
    - 4: "Some students know Java"**

- **Domain:  $U = \{\text{Alice, Bob, Charlie}\}$**
- Let's use:
  - **$F(x)$** : "x is a student"
  - **$J(x)$** : "x knows Java"
  - Solve:
    - 1: "Everything is a student"**
  - This means: **Alice** is a student **AND Bob** is a student **AND Charlie** is a student.
- **Translation:  $\forall x F(x)$**   
*Literally: "For every person x, x is a student."*

- **2: "Nothing is a student"**
- This means: **No one** is a student.
- **Wrong translation:**  $\neg \forall x F(x)$   
*(This means "Not everything is a student"  $\rightarrow$  "At least one is not a student")*
- **Correct translation:**  $\neg \exists x F(x)$   
*"There does NOT exist a person who is a student."*
- **Also correct:**  $\forall x \neg F(x)$   
*"Every person is NOT a student."*
- These two are equivalent:  $\neg \exists x F(x) \equiv \forall x \neg F(x)$



- 3: "All students know Java"

- Translation:  $\forall x (F(x) \rightarrow J(x))$

*Literally: "For every person x, IF x is a student, THEN x knows Java."*

- 4: "Some students know Java"

- Translation:  $\exists x (F(x) \wedge J(x))$

*Literally: "There exists a person x such that x is a student AND x knows Java."*

**NOTE:**

→ **with**  $\forall$  for "all F are..."

$\wedge$  **with**  $\exists$  for "some F are..."

# Exercise

- **Domain:**

- $U = \{\text{Alice, Bob, Charlie, David}\}$

- **Predicates:**

- $P(x)$ : "x plays piano"
- $S(x)$ : "x sings"

- **Statements to Translate and Solve:**

1. "Everyone either plays piano or sings."
2. "Nobody sings."
3. "All piano players sing."
4. "Some people play piano and sing."

| Statement                            | Predicate Logic                                | Literal Translation                                                                |
|--------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------|
| Everyone either plays piano or sings | $\forall x (P(x) \vee S(x))$                   | For every person x, x plays piano or sings                                         |
| Nobody sings**                       | $\forall x \neg S(x)$ or $\neg \exists x S(x)$ | Every person <b>does NOT</b> sing / There <b>does not exist a person</b> who sings |
| All piano players sing               | $\forall x (P(x) \rightarrow S(x))$            | For every person x, if x plays piano, then x sings                                 |
| Some people play piano and sing      | $\exists x (P(x) \wedge S(x))$                 | There exists a person x who plays piano AND sings                                  |

**\*\*WRONG:**  $\neg \forall x S(x) \rightarrow$  "Not everyone sings"  $\rightarrow$  wrong, because it allows some people to sing.